Exercise Set 12.2 — Solutions

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Question 1: True or False?

(i) A parallelogram with a right angle is a rectangle.

TRUE In a parallelogram, opposite angles are equal and consecutive adjacent angles are supplementary (***sum = 180°***). If one interior angle is ***90°***, all four angles are forced to be ***90°***, which satisfies the definition of a rectangle.

(ii) A rhombus with perpendicular diagonals is a square.

FALSE **Every** rhombus naturally possesses diagonals that intersect at perpendicular right angles (***90°***). For a rhombus to be a square, its internal angles must also be right angles (or equivalently, its diagonals must be equal in length).

(iii) If the diagonals of a parallelogram are equal, then it is a rectangle.

TRUE Equal diagonals in a parallelogram imply that adjacent triangles formed by two sides and a diagonal are congruent (by SSS), guaranteeing that all interior angles equal ***90°***.

Question 2

The diagonal *AC* of a parallelogram *ABCD* bisects $\angle A$. Show that it also bisects $\angle C$ and that *ABCD* is a rhombus.

> **Given:** Quadrilateral ***ABCD*** is a parallelogram. Diagonal ***AC*** bisects $\angle A$, so $\angle DAC = \angle BAC$.

Part 1: Show that *AC* bisects $\angle C$

1. Since $AB \parallel DC$ with transversal ***AC***:

$$\angle BAC = \angle DCA \quad \textit{(Alternate Interior Angles)}$$

2. Since $AD \parallel BC$ with transversal ***AC***:

$$\angle DAC = \angle BCA \quad \textit{(Alternate Interior Angles)}$$

3. We are given that $\angle DAC = \angle BAC$. Substituting equal values gives:

$$\angle DCA = \angle BCA$$

Hence, diagonal ***AC*** **bisects** $\angle C$.

Part 2: Show that *ABCD* is a rhombus

1. In triangle ΔABC:

$$\angle BAC = \angle BCA \quad \textit{(since } \angle BAC = \angle DAC = \angle BCA\textit{)}$$

2. Sides opposite to equal angles in a triangle are equal:

$$AB = BC$$

3. In parallelogram $ABCD$, opposite sides are equal ($AB = CD$ and $BC = AD$). Combined with $AB = BC$:

$$AB = BC = CD = DA$$

Since all four sides are equal, $ABCD$ **is a rhombus**.

Question 3: Converses of Properties

Examine the converse statements. Answer with Yes/No and provide the required condition if No.

(i) If the diagonals of a quadrilateral $ABCD$ bisect its angles, must $ABCD$ be a rhombus?

YES If the diagonals bisect all interior angles, opposite sides must be parallel and adjacent sides equal, uniquely defining a rhombus.

(ii) If the diagonals of a parallelogram $ABCD$ bisect each other at right angles, must $ABCD$ be a rhombus?

YES A parallelogram whose diagonals are perpendicular to each other is always a rhombus.

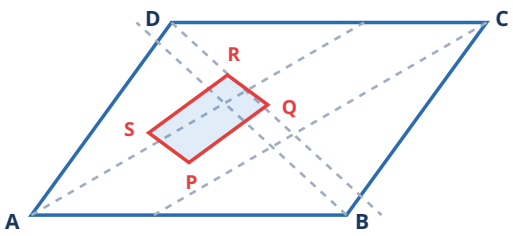
(iii) If the diagonals of a quadrilateral $ABCD$ are of equal length, must $ABCD$ be a rectangle?

NO

Extra Condition: The quadrilateral **must first be a parallelogram** (or its diagonals must bisect each other).
Reason: An isosceles trapezoid has equal diagonals, but it is not a rectangle because it is not a parallelogram.

Question 4

Let $ABCD$ be a parallelogram with $AB \neq BC$. Show that the pairwise intersection points of the four angle bisectors form the vertices of a rectangle. Why did we assume $AB \neq BC$?



Proof:

1. Let the bisectors of angles $\angle A, \angle B, \angle C, \angle D$ intersect pair-wise to form quadrilateral $PQRS$.
2. In ΔAPB, the sum of interior angles is $180°$:

$$\angle APB = 180° - (½ \angle A + ½ \angle B)$$

3. In parallelogram $ABCD$, adjacent angles are supplementary ($\angle A + \angle B = 180°$):

$$½ \angle A + ½ \angle B = ½ (180°) = 90°$$

4. Substituting this into the angle sum equation:

$$\angle APB = 180° - 90° = 90°$$

5. Since $\angle SPQ$ and $\angle APB$ are vertically opposite angles:

$$\angle SPQ = 90°$$

6. By exactly the same reasoning applied to ΔBQC, ΔCRD, and ΔDSA:

$$\angle PQR = 90°, \quad \angle QRS = 90°, \quad \angle RSP = 90°$$

Since all four interior angles of quadrilateral $PQRS$ are right angles ($90°$), **$PQRS$ is a rectangle**.

Why did we assume $AB \neq BC$?

If $AB = BC$, parallelogram $ABCD$ becomes a **rhombus**. In a rhombus, the angle bisectors coincide with the diagonals, which all intersect at a single common central point. Thus, the four bisectors would intersect at one point rather than forming a rectangle (quadrilateral $PQRS$ would degenerate into a single point). Assuming $AB \neq BC$ ensures a non-degenerate rectangle is formed.